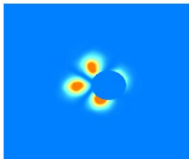
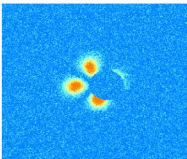
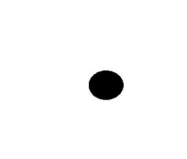
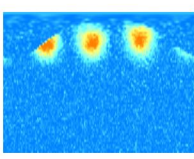
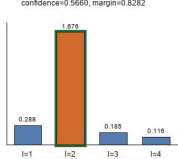
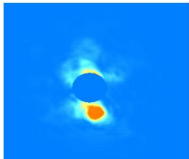
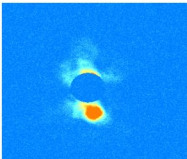
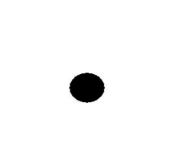
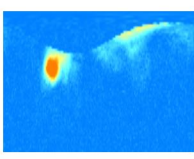
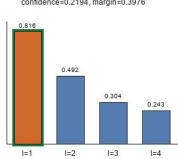
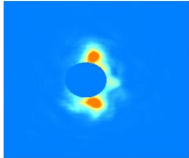
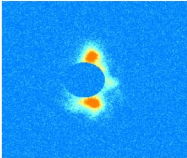
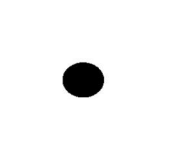
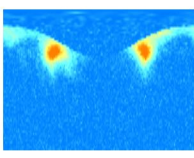
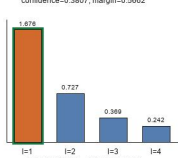
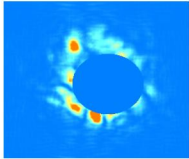
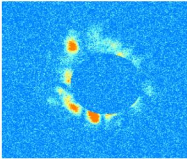
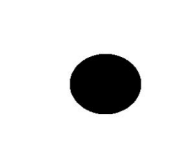
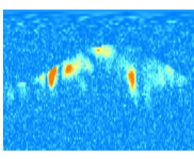
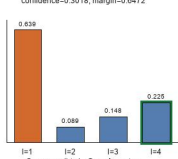


Representative MC-VWLS success and failure cases

Correct: weak turbulence Sample: 21678 True: $l=2$, phase=7 Predicted: $l=2$, phase=7 SNR: 0 dB Occlusion: 0.3 Cn2: 1.00e-14 Distance: 500 m	Occluded clean intensity 	Noisy receiver intensity 	Visible mask 	Polar intensity 	Harmonic scores confidence=0.5660, margin=0.8282  <table><thead><tr><th>Order</th><th>Score</th><th>Type</th></tr></thead><tbody><tr><td>$l=1$</td><td>0.288</td><td>Green frame: true</td></tr><tr><td>$l=2$</td><td>1.678</td><td>Orange: predicted</td></tr><tr><td>$l=3$</td><td>0.185</td><td>Green frame: true</td></tr><tr><td>$l=4$</td><td>0.119</td><td>Green frame: true</td></tr></tbody></table>	Order	Score	Type	$l=1$	0.288	Green frame: true	$l=2$	1.678	Orange: predicted	$l=3$	0.185	Green frame: true	$l=4$	0.119	Green frame: true
Order	Score	Type																		
$l=1$	0.288	Green frame: true																		
$l=2$	1.678	Orange: predicted																		
$l=3$	0.185	Green frame: true																		
$l=4$	0.119	Green frame: true																		
Correct: severe turbulence Sample: 6933 True: $l=1$, phase=4 Predicted: $l=1$, phase=4 SNR: 10 dB Occlusion: 0.3 Cn2: 1.00e-13 Distance: 750 m	Occluded clean intensity 	Noisy receiver intensity 	Visible mask 	Polar intensity 	Harmonic scores confidence=0.2194, margin=0.3976  <table><thead><tr><th>Order</th><th>Score</th><th>Type</th></tr></thead><tbody><tr><td>$l=1$</td><td>0.816</td><td>Orange: predicted</td></tr><tr><td>$l=2$</td><td>0.482</td><td>Green frame: true</td></tr><tr><td>$l=3$</td><td>0.304</td><td>Green frame: true</td></tr><tr><td>$l=4$</td><td>0.243</td><td>Green frame: true</td></tr></tbody></table>	Order	Score	Type	$l=1$	0.816	Orange: predicted	$l=2$	0.482	Green frame: true	$l=3$	0.304	Green frame: true	$l=4$	0.243	Green frame: true
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$l=2$	0.482	Green frame: true																		
$l=3$	0.304	Green frame: true																		
$l=4$	0.243	Green frame: true																		
Failure: adjacent phase Sample: 8158 True: $l=1$, phase=5 Predicted: $l=1$, phase=4 SNR: 5 dB Occlusion: 0.3 Cn2: 5.00e-14 Distance: 1000 m	Occluded clean intensity 	Noisy receiver intensity 	Visible mask 	Polar intensity 	Harmonic scores confidence=0.3807, margin=0.5662  <table><thead><tr><th>Order</th><th>Score</th><th>Type</th></tr></thead><tbody><tr><td>$l=1$</td><td>1.678</td><td>Orange: predicted</td></tr><tr><td>$l=2$</td><td>0.727</td><td>Green frame: true</td></tr><tr><td>$l=3$</td><td>0.389</td><td>Green frame: true</td></tr><tr><td>$l=4$</td><td>0.242</td><td>Green frame: true</td></tr></tbody></table>	Order	Score	Type	$l=1$	1.678	Orange: predicted	$l=2$	0.727	Green frame: true	$l=3$	0.389	Green frame: true	$l=4$	0.242	Green frame: true
Order	Score	Type																		
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$l=2$	0.727	Green frame: true																		
$l=3$	0.389	Green frame: true																		
$l=4$	0.242	Green frame: true																		
Failure: cross order Sample: 37764 True: $l=4$, phase=2 Predicted: $l=1$, phase=4 SNR: 0 dB Occlusion: 0.4 Cn2: 1.00e-13 Distance: 1000 m	Occluded clean intensity 	Noisy receiver intensity 	Visible mask 	Polar intensity 	Harmonic scores confidence=0.3018, margin=0.6472  <table><thead><tr><th>Order</th><th>Score</th><th>Type</th></tr></thead><tbody><tr><td>$l=1$</td><td>0.839</td><td>Orange: predicted</td></tr><tr><td>$l=2$</td><td>0.089</td><td>Green frame: true</td></tr><tr><td>$l=3$</td><td>0.148</td><td>Green frame: true</td></tr><tr><td>$l=4$</td><td>0.225</td><td>Green frame: true</td></tr></tbody></table>	Order	Score	Type	$l=1$	0.839	Orange: predicted	$l=2$	0.089	Green frame: true	$l=3$	0.148	Green frame: true	$l=4$	0.225	Green frame: true
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